Lab 2:

Scenario:
A university is piloting an autonomous delivery drone that transports packages across 6 campus zones (0–5). The drone can only sense its current zone's package status ("Package"/"Empty") and weather status ("Clear"/"Stormy")< it cannot see other zones.

Rules:

1. If the current zone is "Stormy", the drone must hold position (no flying), regardless of package status.
2. If "Clear" and a package is present, the drone should pick up the package.
3. If "Clear" and no package is present, the drone should advance to the next zone.
4. The drone must remember which zones are already confirmed package-free, and report "Route Complete" once all 6 zones are confirmed clear.
5. If the drone holds position at the same zone for more than 3 consecutive steps, it must log a "Reroute Alert" while continuing to hold.

Tasks:

1. Define the PEAS description for the drone.
2. Classify the environment (Observable, Deterministic, Static, Discrete, Agents), with one-sentence justification for each.
3. Design and implement a DroneEnvironment class and a SimpleReflexDroneAgent that follows Rules 1–3 only. Test it on an environment of your own design (6+ zones, at least one Stormy zone) for at least 8 steps.
4. Design and implement a ModelBasedDroneAgent that follows all 5 rules, using an internal state structure of your own design. Run it on the same environment until it reaches "Route Complete".
5. In 4–6 sentences: (a) identify one situation where the two agents would behave differently, and (b) explain precisely why Rule 5 is structurally impossible for a Simple Reflex Agent.